Namit Seth

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Education

Keshav Mahavidyalaya, Delhi University – B.Sc. (H) Electronics Aug 2023 - June 2027

Technical Skills & Concepts

- **Embedded & Hardware**: Microcontrollers (ESP8266, ESP32, ATMega, STM32), RTC (DS1302), Sensor Interfacing, PCB Design, Soldering.
- **Protocols**: I2C, SPI, UART, NEC (IR).
- **Languages & Software**: C/C++ (Embedded), Python, HTML, CSS, JavaScript, SQL, Git, KiCAD, Proteus, LTspice.

Experience

Core Systems Intern, SpaceBorn Jun 2026 – Jul 2026

- Developed foundational embedded firmware on **STM32** using **STM32CubeIDE/CubeMX**, including implementing **I2C/SPI communication wrappers**, sensor data structures, and **UART debugging** within **Git-based workflows**.
- Architected and deployed a custom **MPU6050 driver** for motion tracking, utilizing **register-level manipulation** and **sensor fusion** concepts to process real-time IMU data, while validating system reliability through **simulation environments** like **Wokwi**.

Projects

ATmega32 Development Board (PCB Design / KiCad) https://github.com/namit-seth/ATmega32-Development-Board

- Designed and fabricated a custom ATmega32 development board, from schematic capture to PCB layout.
- Implemented ICSP interfaces for direct microcontroller flashing and designed an op-amp-based 2.5V reference circuit driving a bi-directional LED array to visually debug I/O pin states.

Universal IR Hub (ESP8266 / C++) https://github.com/namit-seth/Universal-IR-hub

- Developed a universal control system by **reverse-engineering** undocumented NEC protocol (0xF7 series) signals, effectively bringing **legacy devices** into the modern **smart home ecosystem** for enhanced automation and interoperability.
- Engineered custom **C++ firmware** to bridge hardware-to-cloud communication, enabling direct, robust integration with **Google Home** via **Sinric Pro** to provide intuitive and reliable remote device control.

DS1302 Embedded Library (C++) https://github.com/namit-seth/DS1302

- Replaced high-overhead standard libraries with a lightweight custom RTC driver authored from scratch using hardware datasheet analysis.
- Implemented bit-banged serial communication and direct register-level control, ensuring stable deployment with a minimal footprint in resource-constrained systems.

Standalone Digital Polling System (Microcontroller) https://github.com/namit-seth/EVM-V2

- Resolved signal bouncing and ghost inputs by engineering software-based debouncing logic, eliminating the need for additional physical hardware capacitors.
- Programmed input-handling logic to detect and reject simultaneous multi-button presses, routing validated totals to a 16×2 LCD interface.

Live Multiplayer Treasure Hunt Game (Flask / SQL) https://github.com/namit-seth/treasure-hunt

- Engineered and deployed a real-time web-based treasure hunt for 10 concurrent teams during a departmental festival.
- Diagnosed critical database locks during simultaneous logins, restoring system stability by restructuring transaction management to resolve race conditions.

Extracurricular Activities

Secretary, Elektronica, Keshav Mahavidyalaya 2025 – 2026

- Directed the core team in organizing Elexonia, the annual electronics festival, managing technical event logistics and cross-functional student teams.

Executive Member, Student Council, Keshav Mahavidyalaya 2025 – 2026

- Fulfilled administrative duties as an appointed representative, acting as an official signatory to validate student council resolutions.